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Large Order-Invariant Bayesian VARs with Stochastic Volatility

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ABSTRACT

Many popular specifications for Vector Autoregressions (VARs) with multivariate stochastic volatility are not invariant to the way the variables are ordered due to the use of a lower triangular parameterization of the error covariance matrix. We show that the order invariance problem in existing approaches is likely to become more serious in large VARs. We propose the use of a specification which avoids the use of this lower triangular parameterization. We show that the presence of multivariate stochastic volatility allows for identification of the proposed model and prove that it is invariant to ordering. We develop a Markov chain Monte Carlo algorithm which allows for Bayesian estimation and prediction. In exercises involving artificial and real macroeconomic data, we demonstrate that the choice of variable ordering can have non-negligible effects on empirical results when using the nonorder invariant approach. In a macroeconomic forecasting exercise involving VARs with 20 variables we find that our order-invariant approach leads to the best forecasts and that some choices of variable ordering can lead to poor forecasts using a conventional, non-order invariant, approach.

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Large vector autoregression; Order invariance; Shrinkage prior; Stochastic volatility

1. Introduction

Vector Autoregressions (VARs) have become one of the most popular models in modern macroeconomics since the seminal work by Sims (1980). In recent years, two prominent developments have occurred in the Baysian VAR literature. First, beginning with Banbura, Giannone, and Reichlin (2010) researchers have been working with large VARs involving dozens or even hundreds of dependent variables.[1] Second, there is an increasing recognition, in most macroeconomic datasets, of the need to allow for time variation in VAR error variances, see, among many others, Clark (2011). Both of these developments have led to VAR specifications which are not invariant to the way the variables are ordered in the VAR. In small VARs, ordering issues can be important, but as we shall show in this article take on additional importance in large VARs.[2] These considerations motivate the present article. In it we describe an alternative VAR specification with stochastic volatility (SV) and prove that it is invariant to the ways the variables are ordered. We develop a computationally-efficient Markov chain Monte Carlo (MCMC) algorithm which is scaleable and allows for Bayesian estimation and forecasting even in high dimensional VARs. We carry out empirical work with artificial and real data which demonstrates the consequences of a lack of order invariance in conventional approaches and that our approach does not suffer from them.

There are two basic insights that underlie our approach. The first is that it is the combination of the use of a lower triangular parameterization of the error covariance matrix $\mathbf{\Sigma}_t$ and the associated priors under this parameterization that lead to order dependence. Beginning with influential early VAR-SV papers such as Cogley and Sargent (2005) and Primiceri (2005), this lower triangular parameterization has been used in numerous papers. However, as noted by Carriero, Clark, and Marcellino (2019), since this multivariate SV specification is constructed based on a lower triangular impact matrix $\mathbf{B}_0$ and priors are independently elicited on $\mathbf{B}_0$ and the SV processes, the implied prior on $\mathbf{\Sigma}_t$ is not order invariant.

The second is that SV can be used to identify the impact matrix $\mathbf{B}_0$, see Bertsche and Braun (2022). We adopt a VAR-SV specification that avoids the use of the lower triangular parameterization. If we were working with a homoscedastic VAR, our model would be unidentified. But we let the SV achieve identification. In theory we prove order invariance and, in practice, we show that it works well.

There are a few recent papers that demonstrate the empirical importance of this ordering issue. For example, Bognanni (2018) and Hartwig (2020) show that results from structural analysis based on the reduced-form VARs of Cogley and Sargent (2005) and Primiceri (2005) can be sensitive to how the variables are ordered. Chan et al. (2018) and Chan (2022) demonstrate that estimates of reduced-form error variances based on the lower triangular parameterization can drastically change across different variable orderings. Finally, Arias, Rubio-Ramirez, and Shin (2022) illustrate the empirical relevance of this ordering issue

[1] In their classic study of the effects of monetary policy, Leeper, Sims, and Zha (1996) estimate a few SVARs of various sizes, including one with 18 variables.

[2] In small VARs, results for a small number of different orderings can be presented as a robustness check. In addition, methods have been proposed for searching over different orderings in papers such as Levy and Lopes (2021) and Wu and Koop (2022). But with a large number of variables a logical ordering of the variables may not present itself and it is not practical to consider every possible ordering.

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for producing density forecasts using all permutations of four macroeconomic variables. The present article contributes to this growing literature but focuses on the implications for high-dimensional models. In particular, we provide some theoretical analysis to show that this ordering issue becomes much worse in high-dimensional settings. More importantly, we propose a new order-invariant model as well as an efficient algorithm which scales well to high-dimensional VARs.

The remainder of this article is organized as follows. The Section 2 discusses ordering issues in VAR-SVs. The Section 3 introduces our multivariate SV process and proves that it is identified and order invariant. The Section 4 adds the VAR component to the model and discusses Bayesian estimation of the resulting order-invariant VAR-SV. The Section 5 uses artificial data to illustrate the performance of our model in terms of accuracy and computation time. The Section 6 investigates the practical consequences of using the non-order invariant approach of Cogley and Sargent (2005) using artificial and real macroeconomic data. Finally, Section 7 carries out a forecasting exercise which demonstrates that our order-invariant approach forecasts well, but that forecasts produced by the model of Cogley and Sargent (2005) are sensitive to the way that the variables are ordered and some ordering choices can lead to poor forecast performance.

2. Ordering Issues in VAR-SVs

Ordering issues in VARs relate to the error covariance matrix, not the VAR coefficients themselves. For instance, Carriero, Clark, and Marcellino (2019) use a VAR-SV which involves a lower triangular parameterization of the error covariance matrix, $\mathbf{\Sigma}_t$. They demonstrate that, conditional on $\mathbf{\Sigma}_t$, the draws of the VAR coefficients are invariant to ordering.[3] Such invariance to ordering of the VAR coefficients holds for all the models discussed in the article. Accordingly, in this section and the next we focus on the multivariate stochastic volatility process. We say a model is invariant to ordering if the order of the variables does not affect the posterior distribution of the error covariance matrix, subject to permutations. In other words, if the ordering of variables i and j in the VAR is changed, then posterior inference on the error covariances and variances associated with variables i and j is unaffected for all i and j.

Before discussing multivariate stochastic volatility approaches which are not order invariant, it is worthwhile noting that there are a few multivariate SV approaches which are order invariant. For example, some multivariate SV models based on Wishart or inverse-Wishart processes are invariant to ordering; examples include Philipov and Glickman (2006), Asai and McAleer (2009), Chan et al. (2018), Shin and Zhong (2020), and Arias, Rubio-Ramirez, and Shin (2022). Instead of using the Wishart or inverse-Wishart processes, Arias, Rubio-Ramirez, and Shin (2022) introduce the DSC-SV-AH model that uses random walk processes to construct a time-varying correlation matrix based on a new parameterization of correlation matrices proposed in Archakov and Hansen (2021). However, estimation of these SV models is computationally intensive as it typically involves drawing from high-dimensional, nonstandard distributions. Consequently, these models are generally not applicable to large datasets. In contrast, the common stochastic volatility models considered in Carriero, Clark, and Marcellino (2016) and Chan (2020) are order invariant and can be estimated quickly even for large systems. However, these models are less flexible as the time-varying error covariance matrix depends on a single SV process (e.g., error variances are always proportional to each other).

Another order invariant approach is based on the discounted Wishart process, which admits efficient filtering and smoothing algorithms for estimation. Multivariate SV models based on this approach are considered in Uhlig (1997), West and Harrison (2006), Prado and West (2010), Bognanni (2018) and Arias, Rubio-Ramirez, and Shin (2022). However, this approach appears to be too tightly parameterized for macroeconomic data and it tends to underperform in terms of both point and density forecasts relative to standard SV models such as Cogley and Sargent (2005) and Primiceri (2005); see, for example, Arias, Rubio-Ramirez, and Shin (2022) for a forecast comparison exercise.

Another model that is potentially order invariant is the factor stochastic volatility model, and recently these models have been applied to large datasets. However, factor models have some well-known drawbacks such as misspecification concerns associated with assuming a high dimensional volatility process is well approximated with a low dimensional set of factors. Kastner (2019) notes that, if one imposes the usual identification restrictions (e.g., factor loadings are triangular), then the model is not order invariant. Kastner (2019) does not impose those restrictions, arguing that identification of the factor loadings is not necessary for many purposes (e.g., forecasting).

The multivariate stochastic volatility specification that is perhaps most commonly used in macroeconomics is that of Cogley and Sargent (2005). The SV specification of the reduced-form errors, the n-dimensional vector $\mathbf{u}_t$, in Cogley and Sargent (2005) can be written as

$$\mathbf{u}_t = \mathbf{B}_0^{-1}\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{D}_t), \tag{1}$$

where $\mathbf{B}_0$ is a unit lower triangular matrix with elements b_{ij} and $\mathbf{D}_t = \text{diag}(e^{h_{1,t}}, \ldots, e^{h_{n,t}})$ is a diagonal matrix consisting of the log-volatility $\mathbf{h}_t = (h_{1,t}, \ldots, h_{n,t})'$. By assuming that $\mathbf{B}_0$ is lower triangular, the implied covariance matrix on $\mathbf{u}_t$ naturally depends on the order of the elements in $\mathbf{u}_t$. In particular, one can show that under standard assumptions, the variance of the ith element of $\mathbf{u}_t$, $u_{i,t}$, increases as i increases. Hence, this ordering issue becomes worse in higher dimensional models. In addition, since the stochastic volatility specification in Primiceri (2005) is based on Cogley and Sargent (2005) but with $\mathbf{B}_0$ time-varying, it has a similar ordering issue.

To illustrate this key point, we assume for simplicity that $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n)$. We further assume that the priors on the lower

[3]It is worth noting that one major reason for triangularizing the system is to allow for equation by equation estimation of the model. Any approach which allows for equation by equation estimation can be shown to lead to large computational benefits since there is no need to manipulate the (potentially enormous) posterior covariance matrix of the VAR coefficients in a single block. Triangularization is the most popular equation by equation method and Carriero, Clark, and Marcellino (2019) show how, relative to full system estimation, triangularization can reduce computational complexity of Bayesian estimation of large VARs from $O(n^6)$ to $O(n^4)$ where n is VAR dimension. This highlights the importance of development of VAR specifications, such as the one in this article, which allow for equation by equation estimation.

triangular elements of $\mathbf{B}_0$ are independent with prior means 0 and variances 1. Since $\mathbf{B}_0\mathbf{u}_t = \boldsymbol{\varepsilon}_t$, we can obtain the elements of $\mathbf{u}_t$ recursively and compute their variances $\text{var}(u_{i,t}) = \mathbb{E}u_{i,t}^2$. In particular, we have

$$\begin{aligned} u_{1,t} &= \varepsilon_{1,t} \\ u_{2,t} &= \varepsilon_{2,t} - b_{2,1}u_{1,t} \\ &\vdots \\ u_{i,t} &= \varepsilon_{i,t} - b_{i,i-1}u_{i-1,t} - \cdots - b_{i,1}u_{1,t}. \end{aligned}$$

Since $\mathbb{E}(b_{i,j}b_{i,k}) = 0$ for all $j \neq k$, the expectation of any cross-product term in $u_{i,t}^2$ is 0. Hence, we have

$$\begin{aligned} \mathbb{E}u_{1,t}^2 &= \mathbb{E}\varepsilon_{1,t}^2 = 1 \\ \mathbb{E}u_{2,t}^2 &= \mathbb{E}\varepsilon_{2,t}^2 + \mathbb{E}b_{2,1}^2\mathbb{E}u_{1,t}^2 = 1 + \mathbb{E}u_{1,t}^2 = 2 \\ &\vdots \\ \mathbb{E}u_{i,t}^2 &= \mathbb{E}\varepsilon_{i,t}^2 + \mathbb{E}b_{i,i-1}^2\mathbb{E}u_{i-1,t}^2 + \cdots + \mathbb{E}b_{i,1}^2\mathbb{E}u_{1,t}^2 = 2^{i-1}. \end{aligned}$$

In other words, the variance of the reduced-form error increases (exponentially) as it is ordered lower in the n-tuple, even though the assumptions about $b_{i,j}$ and $\varepsilon_{i,t}$ are identical across $i = 1, \ldots, n$. In Online Appendix C we provide another illustration using a stochastic volatility specification with a time-varying impact matrix similar to that in Primiceri (2005). There we show that the model has a similar ordering issue.

3. An Order-Invariant Stochastic Volatility Model

3.1. Model Definition

In this section we extend the stochastic volatility specification in Cogley and Sargent (2005) with the goal of constructing an order-invariant model. The key reason why the ordering issue in Cogley and Sargent (2005) occurs is because of the lower triangular assumption for $\mathbf{B}_0$. A simple solution to this problem is to avoid this assumption and assume that $\mathbf{B}_0$ is an unrestricted square matrix. To that end, let $\mathbf{y}_t = (y_{1,t}, \ldots, y_{n,t})'$ be an $n \times 1$ vector of variables that is observed over the periods $t = 1, \ldots, T$. To fix ideas, consider the following model with zero conditional mean:

$$\mathbf{y}_t = \mathbf{B}_0^{-1}\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{D}_t), \tag{2}$$

where $\mathbf{D}_t = \text{diag}(e^{h_{1,t}}, \ldots, e^{h_{n,t}})$ is diagonal. Here we assume that $\mathbf{B}_0$ is nonsingular, but is otherwise unrestricted. Each of the log-volatilities follows a stationary AR(1) process:

$$h_{i,t} = \phi_i h_{i,t-1} + u_{i,t}^h, \quad u_{i,t}^h \sim \mathcal{N}(0, \omega_i^2), \tag{3}$$

for $t = 2, \ldots, T$, where $|\phi_i| < 1$ and the initial condition is specified as $h_{i,1} \sim \mathcal{N}(0, \omega_i^2/(1-\phi_i^2))$. Note that the unconditional mean of the AR(1) process is normalized to be zero for identification. We call this the Order Invariant SV (OI-SV) model.

3.2. Identification and Order-Invariance

The question of identification of our OI-SV model can be dealt with quickly. If there is no SV, it is well-known that $\mathbf{B}_0$ is not identified. In particular, any orthogonal transformation of $\mathbf{B}_0$ gives the same likelihood. But with the presence of SV, Bertsche and Braun (2022) show that $\mathbf{B}_0$ is identified up to permutations and sign changes:

Proposition 1 (Bertsche and Braun (2022)). Consider the stochastic volatility model given in (2)–(3) with $\phi_i \neq 0, i = 1, \ldots, n$. Then, $\mathbf{B}_0$ is unique up to permutation of its column and multiplication of its columns by -1.

The assumptions in Proposition 1 are stronger than necessary for identification. In fact, Bertsche and Braun (2022) show that having $n-1$ SV processes is sufficient for identification and derive results relating to partial identification which occurs if there are $n-r$ SV processes with $r > 1$. Intuitively, allowing for time-varying volatility generates nontrivial autocovariances in the squared reduced form errors. These additional moments help identify $\mathbf{B}_0$. We refer readers to Rigobon (2003), Lanne, Lütkepohl, and Maciejowska (2010) and Lewis (2021) for a more detailed discussion of how identification can be achieved in structural VARs via heteroscedasticity.

It is worth emphasizing that Proposition 1 only guarantees $\mathbf{B}_0$ to be identified up to permutations and sign changes. In the estimation we normalize the signs by restricting the diagonal elements of $\mathbf{B}_0$ to be positive. There are different approaches to deal with the permutation or label-switching problem. Here we assume a weakly informative prior on $\mathbf{B}_0$ (e.g., setting the prior mean of $\mathbf{B}_0$ to be an identity matrix) so that column permutations become less likely. Alternatively, one could postprocess the posterior draws to sort them into the correct categories, as suggested in Kaufmann and Schumacher (2019).

Below we outline a few theoretical properties of the stochastic volatility model described in (2)–(3). First, we show that the model is order invariant, in the sense that if we permute the order of the dependent variables, the likelihood function implied by this new ordering is the same as that of the original ordering, provided that we permute the parameters accordingly. Stacking $\mathbf{y} = (\mathbf{y}_1', \ldots, \mathbf{y}_T')'$ and $\mathbf{h} = (\mathbf{h}_1', \ldots, \mathbf{h}_T')'$, note that the likelihood implied by (2) is

$$\begin{aligned} p(\mathbf{y} \mid \mathbf{B}_0, \mathbf{h}) = {} & (2\pi)^{-\frac{nT}{2}} |\det \mathbf{B}_0|^T \\ & \times \prod_{t=1}^{T} |\det \mathbf{D}_t|^{-\frac{1}{2}} e^{-\frac{1}{2}\sum_{t=1}^{T} \mathbf{y}_t'\mathbf{B}_0'\mathbf{D}_t^{-1}\mathbf{B}_0\mathbf{y}_t}, \end{aligned}$$

where $|\det \mathbf{C}|$ denotes the absolute value of the determinant of the square matrix $\mathbf{C}$. Now, suppose we permute the order of the dependent variables. More precisely, let $\mathbf{P}$ denote an arbitrary permutation matrix of dimension n, and we define $\widetilde{\mathbf{y}}_t = \mathbf{P}\mathbf{y}_t$. By the standard change of variable result, the likelihood of $\widetilde{\mathbf{y}} = (\widetilde{\mathbf{y}}_1', \ldots, \widetilde{\mathbf{y}}_T')'$ is

$$\begin{aligned} & |\det \mathbf{P}^{-1}|^T (2\pi)^{-\frac{nT}{2}} |\det \mathbf{B}_0|^T \prod_{t=1}^{T} | \\ & \qquad \times \det \mathbf{D}_t|^{-\frac{1}{2}} e^{-\frac{1}{2}\sum_{t=1}^{T} (\mathbf{P}^{-1}\widetilde{\mathbf{y}}_t)'\mathbf{B}_0'\mathbf{D}_t^{-1}\mathbf{B}_0\mathbf{P}^{-1}\widetilde{\mathbf{y}}_t} \end{aligned}$$

$$
\begin{aligned}
&= (2\pi)^{-\frac{nT}{2}} |\det \mathbf{B}_0\mathbf{P}'|^T \prod_{t=1}^{T} | \\
&\quad \times \det \mathbf{D}_t|^{-\frac{1}{2}} \mathrm{e}^{-\frac{1}{2}\sum_{t=1}^{T} \widetilde{\mathbf{y}}_t'\mathbf{P}\mathbf{B}_0'\mathbf{D}_t^{-1}\mathbf{B}_0\mathbf{P}'\widetilde{\mathbf{y}}_t} \\
&= (2\pi)^{-\frac{nT}{2}} |\det \mathbf{P}\mathbf{B}_0\mathbf{P}'|^T \prod_{t=1}^{T} | \\
&\quad \times \det \mathbf{P}\mathbf{D}_t\mathbf{P}'|^{-\frac{1}{2}} \mathrm{e}^{-\frac{1}{2}\sum_{t=1}^{T} \widetilde{\mathbf{y}}_t'\mathbf{P}\mathbf{B}_0'\mathbf{P}'\mathbf{P}\mathbf{D}_t^{-1}\mathbf{P}'\mathbf{P}\mathbf{B}_0\mathbf{P}'\widetilde{\mathbf{y}}_t} \\
&= (2\pi)^{-\frac{nT}{2}} |\det \widetilde{\mathbf{B}}_0|^T \prod_{t=1}^{T} | \\
&\quad \times \det \widetilde{\mathbf{D}}_t|^{-\frac{1}{2}} \mathrm{e}^{-\frac{1}{2}\sum_{t=1}^{T} \widetilde{\mathbf{y}}_t'\widetilde{\mathbf{B}}_0'\widetilde{\mathbf{D}}_t^{-1}\widetilde{\mathbf{B}}_0\widetilde{\mathbf{y}}_t} = p(\widetilde{\mathbf{y}} \mid \widetilde{\mathbf{B}}_0, \widetilde{\mathbf{h}}),
\end{aligned}
$$

where $\widetilde{\mathbf{B}}_0 = \mathbf{P}\mathbf{B}_0\mathbf{P}'$, $\widetilde{\mathbf{D}}_t = \mathbf{P}\mathbf{D}_t\mathbf{P}' = \mathrm{diag}(\mathrm{e}^{\widetilde{\mathbf{h}}_t})$ and $\widetilde{\mathbf{h}} = (\widetilde{\mathbf{h}}_1', \ldots, \widetilde{\mathbf{h}}_T')$ with $\widetilde{\mathbf{h}}_t = \mathbf{P}\mathbf{h}_t$. Note that we have used the fact that $\mathbf{P}^{-1} = \mathbf{P}'$ and $|\det \mathbf{P}| = 1$, as all permutation matrices are orthogonal matrices. In addition, since $\mathbf{P}^{-1}\widetilde{\mathbf{y}}_t = \mathbf{y}_t$, the first line in the above derivation is equal to $p(\mathbf{y} \mid \mathbf{B}_0, \mathbf{h})$. Hence, we have shown that $p(\mathbf{y} \mid \mathbf{B}_0, \mathbf{h}) = p(\widetilde{\mathbf{y}} \mid \widetilde{\mathbf{B}}_0, \widetilde{\mathbf{h}})$. We summarize this result in the following proposition.

Proposition 2 (Order Invariance). Let $p(\mathbf{y} \mid \mathbf{B}_0, \mathbf{h})$ denote the likelihood of the stochastic volatility model given in (2). Let $\mathbf{P}$ be an arbitrary $n \times n$ permutation matrix and define $\widetilde{\mathbf{y}}_t = \mathbf{P}\mathbf{y}_t$ and $\widetilde{\mathbf{h}}_t = \mathbf{P}\mathbf{h}_t$. Then, the likelihood with dependent variables $\widetilde{\mathbf{y}}_t$ has exactly the same form but with parameters permuted accordingly. More precisely,

$$p(\mathbf{y} \mid \mathbf{B}_0, \mathbf{h}) = p(\widetilde{\mathbf{y}} \mid \widetilde{\mathbf{B}}_0, \widetilde{\mathbf{h}}),$$

where $\widetilde{\mathbf{B}}_0 = \mathbf{P}\mathbf{B}_0\mathbf{P}'$ and $\widetilde{\mathbf{h}} = (\widetilde{\mathbf{h}}_1', \ldots, \widetilde{\mathbf{h}}_T')$.

Proposition 2 also makes clear why the usual lower triangular assumption on the impact matrix $\mathbf{B}_0$ implies an order-dependent likelihood: to obtain the same likelihood value, the impact matrix corresponding to the permuted vector $\widetilde{\mathbf{y}}$ should be permuted accordingly via $\widetilde{\mathbf{B}}_0 = \mathbf{P}\mathbf{B}_0\mathbf{P}'$. But if $\mathbf{B}_0$ is lower triangular, in general there does not exist a lower triangular $\widetilde{\mathbf{B}}_0$ such that $\widetilde{\mathbf{B}}_0 = \mathbf{P}\mathbf{B}_0\mathbf{P}'$.

Proposition 2 shows that the likelihood conditional on the log-volatility is order invariant. It turns out that one can also recover the unconditional covariance matrix of the permuted vector $\widetilde{\mathbf{y}}_t$ from the unconditional covariance matrix of $\mathbf{y}_t$ by permuting its rows and columns accordingly. To that end, we next derive the unconditional covariance matrix of the data implied by the stochastic volatility model in (2)–(3). More specifically, given the model parameters $\mathbf{B}_0$, ϕ_i and ω_i^2, $i = 1, \ldots, n$, suppose we generate the log-volatility processes via (3). It is straightforward to compute the implied unconditional covariance matrix of $\mathbf{y}_t$.

Since the unconditional distribution of $h_{i,t}$ implied by (3) is $\mathcal{N}(0, \omega_i^2/(1-\phi_i^2))$, the unconditional distribution of $\mathrm{e}^{h_{i,t}}$ is log-normal with mean $\mathrm{e}^{\frac{\omega_i^2}{2(1-\phi_i^2)}}$. Hence, the unconditional mean of $\mathbf{D}_t$ is $\mathbf{V}_\mathbf{D} = \mathrm{diag}(\mathrm{e}^{\frac{\omega_1^2}{2(1-\phi_1^2)}}, \ldots, \mathrm{e}^{\frac{\omega_n^2}{2(1-\phi_n^2)}})$. It then follows that the unconditional covariance matrix of $\mathbf{y}_t$ can be written as

$$\mathbf{\Sigma} \equiv (\mathbf{B}_0'\mathbf{V}_\mathbf{D}^{-1}\mathbf{B}_0)^{-1} = (\overline{\mathbf{B}}_0'\overline{\mathbf{B}}_0)^{-1},$$

where $\overline{\mathbf{B}}_0 = \mathbf{V}_\mathbf{D}^{-\frac{1}{2}}\mathbf{B}_0$, that is, scaling each row of $\mathbf{B}_0$ by $\mathrm{e}^{-\frac{\omega_i^2}{4(1-\phi_i^2)}}$, $i = 1, \ldots, n$. From this expression it is also clear that if we permute the order of the dependent variables via $\widetilde{\mathbf{y}}_t = \mathbf{P}\mathbf{y}_t$, the unconditional covariance matrix of $\widetilde{\mathbf{y}}_t$ can be obtained by permuting the rows and columns of $\overline{\mathbf{B}}_0$ accordingly. More precisely:

$$\widetilde{\mathbf{\Sigma}} \equiv \mathbf{P}\mathbf{\Sigma}\mathbf{P}' = (\mathbf{P}\overline{\mathbf{B}}_0'\mathbf{P}'\mathbf{P}\overline{\mathbf{B}}_0\mathbf{P}')^{-1} = (\widetilde{\overline{\mathbf{B}}}_0'\widetilde{\overline{\mathbf{B}}}_0)^{-1},$$

where $\widetilde{\overline{\mathbf{B}}}_0 = \mathbf{P}\overline{\mathbf{B}}_0\mathbf{P}'$. Hence, the unconditional variance of any element in $\mathbf{y}_t$ does not depend on its position in the n-tuple.

This establishes order-invariance in the likelihood function defined by the OI-SV. Of course, the posterior will also be order invariant (subject to permutations and sign switches) if the prior is. This will hold in any reasonable case. All it requires is that the prior for the parameters in equation i under one ordering becomes the prior for equation j in a different ordering if variable i in the first ordering becomes variable j in the second ordering.

4. A VAR with an Order-Invariant SV Specification

4.1. Model Definition

We now add the VAR part to our OI-SV model, leading to the OI-VAR-SV which is given by

$$\mathbf{y}_t = \mathbf{a} + \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{B}_0^{-1}\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{D}_t), \quad (4)$$

where $\mathbf{a}$ is an $n \times 1$ vector of intercepts, $\mathbf{A}_1, \ldots, \mathbf{A}_p$ are $n \times n$ matrices of VAR coefficients, $\mathbf{B}_0$ is nonsingular but otherwise unrestricted $n \times n$ matrix, and $\mathbf{D}_t = \mathrm{diag}(\mathrm{e}^{h_{1,t}}, \ldots, \mathrm{e}^{h_{n,t}})$ is diagonal.[4] Finally, each of the log-volatility $h_{i,t}$ follows the stationary AR(1) process as specified in (3).

4.2. Shrinkage Priors

We argued previously that ordering issues are likely to be most important in high dimensional models. With large VARs over-parameterization concerns can be substantial and, for this reason, Bayesian methods involving shrinkage priors are commonly used. In this section, we describe a particular VAR prior with attractive properties. But it is worth noting that any of the standard Bayesian VAR priors (e.g., the Minnesota prior) could have been used and the resulting model would still have been order invariant.

Let $\boldsymbol{\alpha}_i$ denote the VAR coefficients in the ith equation, $i = 1, \ldots, n$. We consider the Minnesota-type adaptive hierarchical priors proposed in Chan (2021), which have the advantages of both the Minnesota priors (e.g., rich prior beliefs such as cross-variable shrinkage) and modern adaptive hierarchical priors (e.g., heavy-tails and substantial mass around 0). In particular, we construct a Minnesota-type horseshoe prior as follows. For $\alpha_{i,j}$, the jth coefficient in the ith equation, let $\kappa_{i,j} = \kappa_1$ if it is a coefficient on an "own lag" and let $\kappa_{i,j} = \kappa_2$ if it is a coefficient on an "other lag." Given the constants $C_{i,j}$ defined below, consider the horseshoe prior on the VAR coefficients

[4]Following the bulk of the literature on large Bayesian VARs, we focus on the case of constant VAR coefficients. The model can be readily extended to accommodate time-varying VAR coefficients, but with additional computational costs.

(excluding the intercepts) $\alpha_{i,j}, i = 1, \ldots, n, j = 2, \ldots, k$ with $k = np + 1$:

$$(\alpha_{i,j} \mid \kappa_1, \kappa_2, \psi_{i,j}) \sim \mathcal{N}(m_{i,j}, \kappa_{i,j}\psi_{i,j}C_{i,j}), \tag{5}$$

$$\sqrt{\psi}_{i,j} \sim \mathcal{C}^+(0,1), \tag{6}$$

$$\sqrt{\kappa}_1, \sqrt{\kappa}_2 \sim \mathcal{C}^+(0,1), \tag{7}$$

where $\mathcal{C}^+(0,1)$ denotes the standard half-Cauchy distribution. κ_1 and κ_2 are the global variance components that are common to, respectively, coefficients of own and other lags, whereas each $\psi_{i,j}$ is a local variance component specific to the coefficient $\alpha_{i,j}$. Lastly, the constants $C_{i,j}$ are obtained as in the Minnesota prior, that is,

$$C_{i,j} = \begin{cases} \frac{1}{l^2}, & \text{for the coefficient on the } l\text{th lag of variable } i, \\ \frac{s_i^2}{l^2 s_j^2}, & \text{for the coefficient on the } l\text{th lag of variable} \\ & j,\ j \neq i, \end{cases} \tag{8}$$

where s_r^2 denotes the sample variance of the residuals from an AR(4) model for the variable $r, r = 1, \ldots, n$. Furthermore, for data in growth rates $m_{i,j}$ is set to be 0; for level data, $m_{i,j}$ is set to be zero as well except for the coefficient associated with the first own lag, which is set to be one.

It is easy to verify that if all the local variances are fixed, that is, $\psi_{i,j} \equiv 1$, then the Minnesota-type horseshoe prior given in (5)–(7) reduces to the Minnesota prior.[5] Hence, it can be viewed as an extension of the Minnesota prior by introducing a local variance component such that the marginal prior distribution of $\alpha_{i,j}$ has heavy tails. On the other hand, if $m_{i,j} = 0$, $C_{i,j} = 1$ and $\kappa_1 = \kappa_2$, then the Minnesota-type horseshoe prior reduces to the standard horseshoe prior where the coefficients have identical distributions. Therefore, it can also be viewed as an extension of the horseshoe prior (Carvalho, Polson, and Scott 2010) that incorporates richer prior beliefs on the VAR coefficients, such as cross-variable shrinkage, that is, shrinking coefficients on own lags differently than other lags.

To facilitate estimation, we follow Makalic and Schmidt (2016) and use the following latent variable representations of the half-Cauchy distributions in (6)–(7):

$$(\psi_{i,j} \mid z_{\psi_{i,j}}) \sim \mathcal{IG}(1/2, 1/z_{\psi_{i,j}}), \quad z_{\psi_{i,j}} \sim \mathcal{IG}(1/2, 1), \tag{9}$$

$$(\kappa_l \mid z_{\kappa_l}) \sim \mathcal{IG}(1/2, 1/z_{\kappa_l}), \quad z_{\kappa_l} \sim \mathcal{IG}(1/2, 1), \tag{10}$$

for $i = 1, \ldots, n, j = 2, \ldots, k$ and $l = 1, 2$, where $\mathcal{IG}(\alpha, \beta)$ denotes the inverse gamma distribution with density $p(x) = \beta^\alpha \Gamma(\alpha)^{-1} x^{-(\alpha+1)} \mathrm{e}^{-\beta/x}$.

Next, we specify the priors on other model parameters. Let $\mathbf{b}_i$ denote the ith row of $\mathbf{B}_0$ for $i = 1, \ldots, n$, that is, $\mathbf{B}_0' = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$. We consider independent priors on $\mathbf{b}_i, i = 1, \ldots, n$:[6]

$$\mathbf{b}_i \sim \mathcal{N}(\mathbf{b}_{0,i}, \mathbf{V}_{\mathbf{b}_i}).$$

For the parameters in the stochastic volatility equations, we assume the priors for $j = 1, \ldots, n$:

$$\phi_j \sim \mathcal{N}(\phi_{0,j}, V_{\phi_j})1(|\phi_j| < 1), \quad \sigma_j^2 \sim \mathcal{IG}(\nu_j, S_j).$$

4.3. MCMC Algorithm

In this section we develop a posterior sampler which allows for Bayesian estimation of the order-invariant stochastic volatility model with the Minnesota-type horseshoe prior. For later reference, let $\boldsymbol{\psi}_i = (\psi_{i,1}, \ldots, \psi_{i,k})'$ denote the local variance components associated with $\boldsymbol{\alpha}_i$ and define $\boldsymbol{\kappa} = (\kappa_1, \kappa_2)'$. Furthermore, let $\mathbf{z}_{\boldsymbol{\psi}_i} = (z_{\psi_{i,1}}, \ldots, z_{\psi_{i,k}})'$ denote the latent variables corresponding to $\boldsymbol{\psi}_i$ and similarly define $\mathbf{z}_{\boldsymbol{\kappa}}$. Next, stack $\mathbf{y} = (\mathbf{y}_1', \ldots, \mathbf{y}_T')'$, $\boldsymbol{\alpha} = (\boldsymbol{\alpha}_1', \ldots, \boldsymbol{\alpha}_n')'$, $\boldsymbol{\psi} = (\boldsymbol{\psi}_1', \ldots, \boldsymbol{\psi}_n')'$, $\mathbf{z}_{\boldsymbol{\psi}} = (\mathbf{z}_{\boldsymbol{\psi}_1}', \ldots, \mathbf{z}_{\boldsymbol{\psi}_n}')'$ and $\mathbf{h} = (\mathbf{h}_1', \ldots, \mathbf{h}_T')'$ with $\mathbf{h}_t = (h_{1,t}, \ldots, h_{n,t})'$. Finally, let $\mathbf{h}_{i,\cdot} = (h_{i,1}, \ldots, h_{i,T})'$ represent the vector of log-volatility for the ith equation, $i = 1, \ldots, n$. Then, posterior draws can be obtained by sampling sequentially from each of the full conditional distributions. Below we discuss sampling of $\mathbf{B}_0$ from $p(\mathbf{B}_0 \mid \mathbf{y}, \boldsymbol{\alpha}, \mathbf{h}, \boldsymbol{\phi}, \boldsymbol{\omega}^2, \boldsymbol{\psi}, \boldsymbol{\kappa}, \mathbf{z}_{\boldsymbol{\psi}}, \mathbf{z}_{\boldsymbol{\kappa}}) = p(\mathbf{B}_0 \mid \mathbf{y}, \boldsymbol{\alpha}, \mathbf{h})$. The details of the rest of the posterior sampler are given in Online Appendix B. In addition, we report in Online Appendix C the inefficiency factors of the MCMC samples associated with the empirical application. The results indicates that the posterior sampler is efficient in terms of being able to generate MCMC draws with relatively low autocorrelations.

To sample $\mathbf{B}_0$, we adapt the sampling approach in Waggoner and Zha (2003) and Villani (2009) to the setting with stochastic volatility. More specifically, we aim to draw each row of $\mathbf{B}_0$ given all other rows. To that end, we first rewrite (4) as

$$(\mathbf{Y} - \mathbf{XA})\mathbf{B}_0' = \mathbf{E}, \tag{11}$$

where $\mathbf{Y}$ is the $T \times n$ matrix of dependent variables, $\mathbf{X}$ is the $T \times k$ matrix of lagged dependent variables with $k = 1 + np$, $\mathbf{A} = (\mathbf{a}, \mathbf{A}_1, \ldots, \mathbf{A}_n)'$ is the $k \times n$ matrix of VAR coefficients and $\mathbf{E}$ is the $T \times n$ matrix of errors. Then, for $i = 1, \ldots, n$, we have

$$(\mathbf{Y} - \mathbf{XA})\mathbf{b}_i = \mathbf{E}_i, \quad \mathbf{E}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}_{\mathbf{h}_{i,\cdot}}),$$

where $\mathbf{E}_i$ is the ith column of $\mathbf{E}$ and $\boldsymbol{\Omega}_{\mathbf{h}_{i,\cdot}} = \mathrm{diag}(\mathrm{e}^{h_{i,1}}, \ldots, \mathrm{e}^{h_{i,T}})$. Hence, the full conditional distribution of $\mathbf{b}_i$ is given by

$$\begin{aligned} p(\mathbf{b}_i \mid \mathbf{y}, \boldsymbol{\alpha}, \mathbf{b}_{-i}, \mathbf{h}_{i,\cdot}) &\propto |\det \mathbf{B}_0|^T \mathrm{e}^{-\frac{1}{2}\mathbf{b}_i'(\mathbf{Y}-\mathbf{XA})'\boldsymbol{\Omega}_{\mathbf{h}_{i,\cdot}}^{-1}(\mathbf{Y}-\mathbf{XA})\mathbf{b}_i} \\ &\quad \times \mathrm{e}^{-\frac{1}{2}(\mathbf{b}_i-\mathbf{b}_{0,i})'\mathbf{V}_{\mathbf{b}_i}^{-1}(\mathbf{b}_i-\mathbf{b}_{0,i})} \\ &\propto |\det \mathbf{B}_0|^T \mathrm{e}^{-\frac{1}{2}(\mathbf{b}_i-\widehat{\mathbf{b}}_i)'\mathbf{K}_{\mathbf{b}_i}(\mathbf{b}_i-\widehat{\mathbf{b}}_i)} \end{aligned} \tag{12}$$

where

$$\mathbf{K}_{\mathbf{b}_i} = \mathbf{V}_{\mathbf{b}_i}^{-1} + (\mathbf{Y} - \mathbf{XA})'\boldsymbol{\Omega}_{\mathbf{h}_{i,\cdot}}^{-1}(\mathbf{Y} - \mathbf{XA}), \quad \widehat{\mathbf{b}}_i = \mathbf{K}_{\mathbf{b}_i}^{-1}\mathbf{V}_{\mathbf{b}_i}^{-1}\mathbf{b}_{0,i}.$$

The above full conditional posterior distribution is nonstandard and direct sampling is infeasible. Fortunately, Waggoner and Zha (2003) develop an efficient algorithm to sample $\mathbf{b}_i$ in the case when $\widehat{\mathbf{b}}_i = \mathbf{0}$. Villani (2009) further generalizes this sampling approach for nonzero $\widehat{\mathbf{b}}_i$.

[5] There are many different versions of "the Minnesota prior." The version that we are using is an independent normal and inverse-Wishart prior that is applicable to VARs with time-varying volatility. See, for example, eq. (14) in Karlsson (2013) or eq. (13) in Carriero, Clark, and Marcellino (2015).

[6] In the simulations and the empirical application, we set $\mathbf{b}_{0,i} = \mathbf{e}_i$ and $\mathbf{V}_{\mathbf{b}_i} = \mathbf{I}_n$, where $\mathbf{e}_i$ is the ith column of the identity matrix $\mathbf{I}_n$. This choice of hyperparameters implies an order invariant prior in the sense that if we permute the endogenous variables $\widetilde{\mathbf{y}}_t = \mathbf{P}\mathbf{y}_t$, the prior mean of $\mathbf{P}\mathbf{B}_0\mathbf{P}'$ is also an identity matrix (and all prior variances of the elements of $\mathbf{P}\mathbf{B}_0\mathbf{P}'$ are also 1). Since $\mathbf{B}_0$ is identified up to permutations and sign normalization, this relatively weak prior works well in a range of simulations.

The key idea is to show that $\mathbf{b}_i$ has the same distribution as a linear transformation of an n-vector that consists of one absolute normal and $n-1$ normal random variables. A random variable Z follows the absolute normal distribution $\mathcal{AN}(\mu,\rho)$ if it has the density function

$$f_{\mathcal{AN}}(z;\mu,\rho) = c|z|^{\frac{1}{\rho}} e^{-\frac{1}{2\rho}(z-\mu)^2}, \quad z\in\mathbb{R}, \rho\in\mathbb{R}^+, \mu\in\mathbb{R},$$

where c is a normalizing constant.

To formally state the results, let $\mathbf{C}_i$ denote the Cholesky factor of $T^{-1}\mathbf{K}_{\mathbf{b}_i}$ such that $\mathbf{K}_{\mathbf{b}_i} = T\mathbf{C}_i\mathbf{C}_i'$. Let $\mathbf{F}_{-i}$ represent the matrix $\mathbf{F}$ with the ith column deleted and $\mathbf{F}^{\perp}$ be the orthogonal complement of $\mathbf{F}$. Furthermore, define $\mathbf{v}_1 = \mathbf{C}_i^{-1}(\mathbf{B}_{0,-i}')^{\perp}/||\mathbf{C}_i^{-1}(\mathbf{B}_{0,-i}')^{\perp}||$, where $(\mathbf{B}_{0,-i}')^{\perp} \equiv ((\mathbf{B}_0')_{-i})^{\perp}$, and let $(\mathbf{v}_2,\ldots,\mathbf{v}_n) = \mathbf{v}_1^{\perp}$. Then, we claim that

$$(\mathbf{b}_i \mid \mathbf{y},\boldsymbol{\alpha},\mathbf{b}_{-i},\mathbf{h}_{i,\cdot}) \overset{\mathrm{d}}{=} (\mathbf{C}_i')^{-1}\sum_{j=1}^{n}\xi_j\mathbf{v}_j,$$

where $\xi_1 \sim \mathcal{AN}(\widehat{\xi}_1, T^{-1})$ and $\xi_j \sim \mathcal{N}(\widehat{\xi}_j, T^{-1}), j=2,\ldots,n$ with $\widehat{\xi}_j = \widehat{\mathbf{b}}_i'\mathbf{C}_i\mathbf{v}_j$.

To prove the claim, let $\mathbf{b}_i = (\mathbf{C}_i')^{-1}\sum_{j=1}^{n}\xi_j\mathbf{v}_j$. It suffices to show that if we substitute $\mathbf{b}_i$ into (12), $\xi_1,\ldots,\xi_n$ are independent with $\xi_1 \sim \mathcal{AN}(\widehat{\xi}_1, T^{-1})$ and $\xi_j \sim \mathcal{N}(\widehat{\xi}_j, T^{-1}), j=2,\ldots,n$. To that end, note that by construction, $\mathbf{v}_1,\ldots,\mathbf{v}_n$ forms an orthonormal basis of $\mathbb{R}^n$, particularly $\sum_{j=1}^{n}\mathbf{v}_j\mathbf{v}_j' = \mathbf{I}_n$. We first write the quadratic form in (12) as

$$\begin{aligned}(\mathbf{b}_i-\widehat{\mathbf{b}}_i)'\mathbf{K}_{\mathbf{b}_i}(\mathbf{b}_i-\widehat{\mathbf{b}}_i) &= T\left(\sum_{j=1}^{n}\xi_j\mathbf{v}_j - \mathbf{C}_i'\widehat{\mathbf{b}}_i\right)'\left(\sum_{j=1}^{n}\xi_j\mathbf{v}_j - \mathbf{C}_i'\widehat{\mathbf{b}}_i\right)\\
&= T\left(\sum_{j=1}^{n}\xi_j^2 - 2\sum_{j=1}^{n}\xi_j\widehat{\mathbf{b}}_i'\mathbf{C}_i\mathbf{v}_j + \widehat{\mathbf{b}}_i'\mathbf{C}_i\left(\sum_{j=1}^{n}\mathbf{v}_j\mathbf{v}_j'\right)\mathbf{C}_i\widehat{\mathbf{b}}_i\right)\\
&= T\sum_{j=1}^{n}(\xi_j-\widehat{\xi}_j)^2.\end{aligned}$$

Next, note that by construction, $(\mathbf{v}_2,\ldots,\mathbf{v}_n)$ spans the same space as $\mathbf{B}_{0,-i}'$. Hence, it follows that

$$\begin{aligned}|\det\mathbf{B}_0| = |\det\mathbf{B}_0'| = |\det(\mathbf{b}_1,\ldots,\mathbf{b}_{i-1},(\mathbf{C}_i')^{-1}\\ \times\sum_{j=1}^{n}\xi_j\mathbf{v}_j,\mathbf{b}_{i+1},\ldots,\mathbf{b}_n)| \propto |\xi_1|.\end{aligned}$$

Finally, substituting the quadratic form and the determinant into (12), we obtain

$$\begin{aligned}p(\mathbf{b}_i \mid \mathbf{y},\boldsymbol{\alpha},\mathbf{b}_{-i},\mathbf{h}_{i,\cdot}) &\propto |\xi_1|^T e^{-\frac{T}{2}\sum_{j=1}^{n}(\xi_j-\widehat{\xi}_j)^2}\\
&= |\xi_1|^T e^{-\frac{T}{2}(\xi_1-\widehat{\xi}_1)^2}\prod_{j=2}^{n} e^{-\frac{T}{2}(\xi_j-\widehat{\xi}_j)^2}.\end{aligned}$$

In other words, $\xi_1 \sim \mathcal{AN}(\widehat{\xi}_1, T^{-1})$ and $\xi_j \sim \mathcal{N}(\widehat{\xi}_j, T^{-1})$, $j=2,\ldots,n$, and we have proved the claim.

In order to use the above result, we need an efficient way to sample from $\mathcal{AN}(\widehat{\xi}_1, T^{-1})$. This can be done by using the two-component normal mixture approximation considered in Appendix C of Villani (2009). In addition, the orthogonal complement of $\mathbf{v}_1$, namely, $\mathbf{v}_1^{\perp}$, can be obtained using the singular value decomposition.[7] In the sampler we fix the sign of the ith element of $\mathbf{b}_i$ to be positive (so that the diagonal elements of $\mathbf{B}_0$ are positive). This is done by multiplying the draw $\mathbf{b}_i$ by -1 if its ith element is negative. Details of the other steps of the posterior sampler are given in Online Appendix B.

5. Experiments with Artificial Data

In this section we first present results on two artificial data experiments to illustrate how the new model performs under DGPs with and without stochastic volatility. In the first experiment, we generate a dataset from the VAR in (4) with $n=3$, $T=500$, $p=4$ and the stochastic volatility processes as specified in (3). We then estimate the model using the posterior sampler outlined in the previous section. The first dataset is generated as follows. First, the intercepts are drawn independently from $\mathcal{U}(-10,10)$, the uniform distribution on the interval $(-10,10)$. For the VAR coefficients, the diagonal elements of the first VAR coefficient matrix are iid $\mathcal{U}(0,0.5)$ and the off-diagonal elements are from $\mathcal{U}(-0.2,0.2)$; all other elements of the jth ($j>1$) VAR coefficient matrices are iid $\mathcal{N}(0,0.1^2/j^2)$. For the impact matrix $\mathbf{B}_0$, the diagonal elements are iid $\mathcal{U}(0.5,2)$, whereas the off-diagonal elements are iid $\mathcal{N}(0,1)$. Finally, for the SV processes, we set $\phi_i = 0.95$ and $\omega_i^2 = 0.05, i=1,\ldots,n$. The results of the artificial data experiments are reported in Figures 1–3.

Overall, all the estimates track the true values closely. In particular, the new model is able to recover the time-varying reduced-form covariance matrices. In addition, it is also evident that $\mathbf{B}_0$ can be estimated accurately.

In the second experiment, we generate data from the same VAR but with the stochastic volatility component turned off so that the errors are homoscedastic (in particular, we set $h_{i,t}=0$). In other words, we have generated data from a homoscedastic, unidentified model. But we are estimating it using the (misspecified) heteroscedastic model with the posterior sampler outlined in the previous sections. The results are reported in Figures 4–6.

When the DGP does not have stochastic volatility, elements of the impact matrix $\mathbf{B}_0$ are much harder to pin down (as expected since these parameters are not identified). But it is interesting to note that the estimates of the time-varying variances are still able to track the true values fairly closely (as these reduced-form error variances are identified). The VAR coefficients, as well, are well-estimated.

Finally, we document the runtimes of estimating OI-VAR-SV models of different dimensions to assess how well the posterior sampler scales to higher dimensions. More specifically, we report in Table 1 the computation times (in minutes) to obtain 10,000 posterior draws from the proposed OI-VAR-SV model of dimensions $n=10,20,50$, and sample sizes $T=300,800$. The posterior sampler is implemented in MATLAB on a desktop with an Intel Core i7-7700 @3.60 GHz processor and 64 GB memory. It is evident from the table that for typical applications with

[7]In MATLAB, the orthogonal complement of $\mathbf{v}_1$ can be obtained using `null`($\mathbf{v}_1'$).

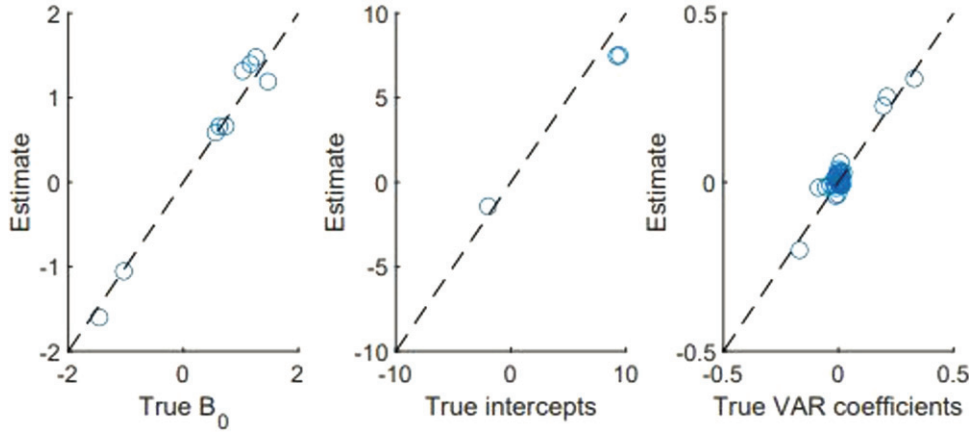

Figure 1. Scatterplot of the posterior means of $\mathbf{B}_0$ (left panel), intercepts (middle panel), and VAR coefficients (right panel) against true values from a DGP with stochastic volatility.

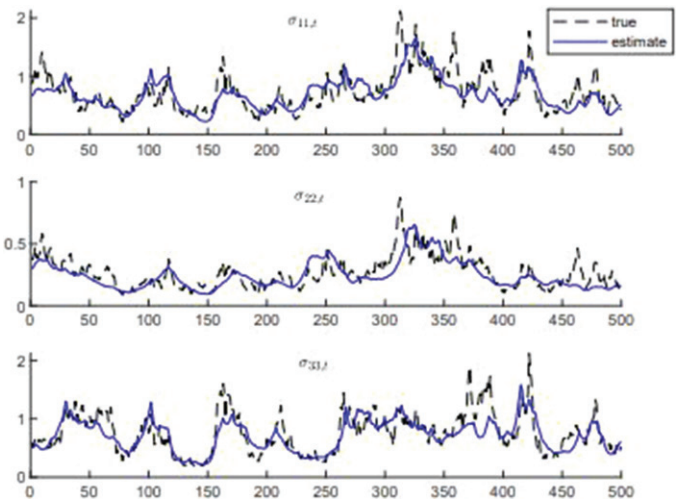

Figure 2. Estimated time-varying reduced-form variances, where $\sigma_{ii,t}$ denote the ith diagonal element of $\mathbf{\Sigma}_t = \mathbf{B}_0^{-1}\mathbf{D}_t(\mathbf{B}_0^{-1})'$, from a DGP with stochastic volatility.

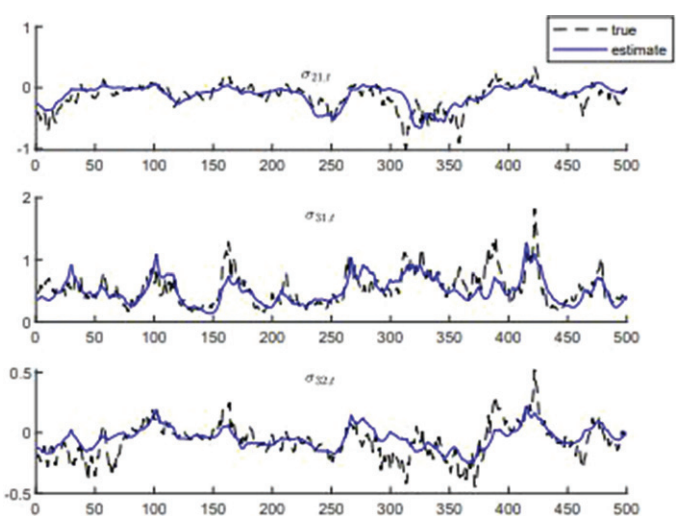

Figure 3. Estimated time-varying reduced-form covariances, where $\sigma_{ij,t}$ denote the (i,j) element of $\mathbf{\Sigma}_t = \mathbf{B}_0^{-1}\mathbf{D}_t(\mathbf{B}_0^{-1})'$, from a DGP with stochastic volatility.

15–30 variables, the OI-VAR-SV model can be estimated quickly. More generally, its estimation time is comparable to that of the triangular model in Carriero, Clark, and Marcellino (2019).

6. Sensitivity to Ordering in the VAR-SV of Cogley and Sargent (2005)

In the preceding sections, we have developed Bayesian methods for order-invariant inference in VARs with SV. In this section, we illustrate the importance of this by showing the degree to which results are sensitive to ordering in the VAR-SV of Cogley and Sargent (2005), hereafter CS-VAR-SV, which is one of the most popular VAR-SV specifications in empirical macroeconomics.[8] We do this using both simulated and macroeconomic data. To ensure comparability, we use the same prior for the CS-VAR-SV as for our OI-VAR-SV. In particular, we set the hyperparameters $\phi_{0,j} = 0.95$, $V_{\phi_j} = 0.05^2$, $\nu_j = 3$ and $S_j = 0.1, j = 1, \ldots, n$, for both models. The only exception is the prior on $\mathbf{B}_0$. For the CS-VAR-SV, $\mathbf{B}_0$ is unit lower triangular, and we assume the lower triangular elements have prior distribution $\mathcal{N}(0,1)$. For OI-VAR-SV, we assume the same $\mathcal{N}(0,1)$ prior for the off-diagonal elements of $\mathbf{B}_0$, whereas the diagonal elements are iid $\mathcal{N}(1,1)$. MCMC estimation of the CS-VAR-SV is carried out using the algorithm of Carriero et al. (2022).

6.1. A Simulation Experiment

As discussed in Primiceri (2005), the lower triangular parameterization used in the model of Cogley and Sargent (2005) is unable to accommodate certain co-volatility patterns. To illustrate this point, we simulate $T = 500$ observations from the model in (3) and (4) with $p = 4$ and $n = 3$. To investigate the consequences of the lower triangular restriction we choose a non-triangular data generating process (DGP). We emphasise that the presence of SV in the DGP means that the model is identified. We set

$$\mathbf{B}_0 = \begin{pmatrix} 1 & -0.8 & -0.8 \\ 0.8 & 1 & -0.8 \\ 0.8 & 0.8 & 1 \end{pmatrix}.$$

The VAR coefficients and volatilities are simulated in exactly the same way as described in Section 5. Figure 7 reports the

[8]In the original model of Cogley and Sargent (2005), the log-volatility follows a unit root process. Here we consider a version in which the log-volatility follows a stationary AR(1) process (with a nonzero mean) so that it is directly comparable to the OI-VAR-SV.

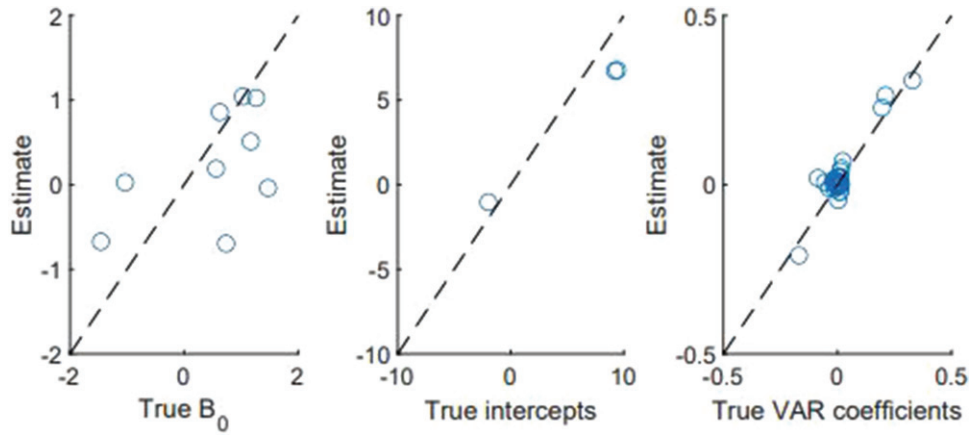

Figure 4. Scatterplot of the posterior means of $\mathbf{B}_0$ (left panel), intercepts (middle panel), and VAR coefficients (right panel) against true values from a DGP without stochastic volatility.

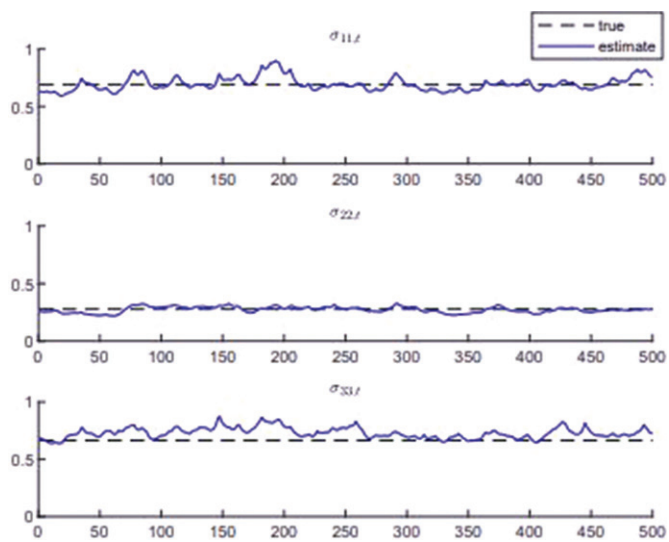

Figure 5. Estimated time-varying reduced-form variances, where $\sigma_{ii,t}$ denote the ith diagonal element of $\mathbf{\Sigma}_t = \mathbf{B}_0^{-1}\mathbf{D}_t(\mathbf{B}_0^{-1})'$, from a DGP without stochastic volatility.

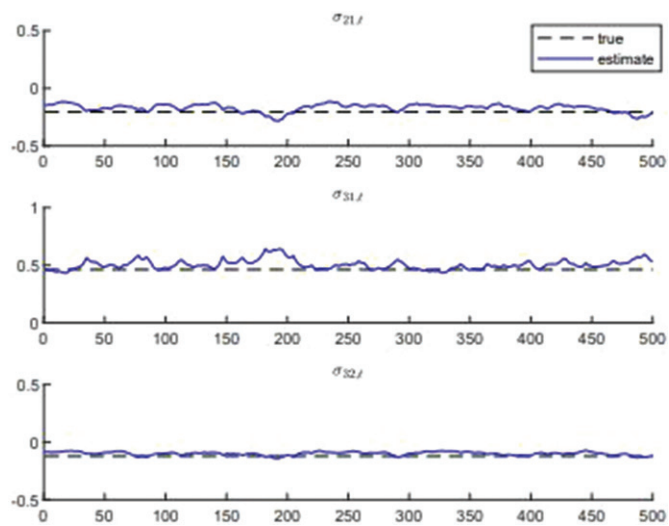

Figure 6. Estimated time-varying reduced-form covariances, where $\sigma_{ij,t}$ denote the (i,j) element of $\mathbf{\Sigma}_t = \mathbf{B}_0^{-1}\mathbf{D}_t(\mathbf{B}_0^{-1})'$, from a DGP without stochastic volatility.

Table 1. The computation times (in minutes) to obtain 10,000 posterior draws from the proposed order invariant VAR-SV model with n variables and T observations.

$T = 300$			$T = 800$		
$n = 10$	$n = 20$	$n = 50$	$n = 10$	$n = 20$	$n = 50$
2.5	17.5	233	6.8	39.7	630

NOTE: All VARs have $p = 4$ lags.

estimated time-varying reduced-form error covariance matrices produced by the CS-VAR-SV and OI-VAR-SV models. It is clear from the figure that the estimates from the CS-VAR-SV model tend to be flat and do not capture the time-variation in the error covariances well. This is perhaps not surprising as the $\mathbf{B}_0$ is restricted to be lower triangular in the CS-VAR-SV. However, the OI-VAR-SV does track the true $\sigma_{ij,t}$ quite well. Since error covariances play an important role in features of interest such as impulse responses, this illustrates the potential negative consequences of working with triangularized systems.

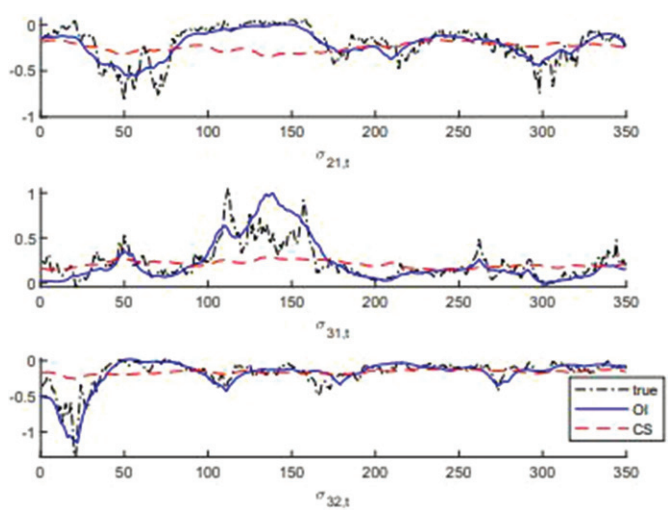

Figure 7. Estimated time-varying reduced-form covariances from the OI-VAR-SV and CS-VAR-SV models, where $\sigma_{ij,t}$ denote the (i,j) element of $\mathbf{\Sigma}_t = \mathbf{B}_0^{-1}\mathbf{D}_t(\mathbf{B}_0^{-1})'$.

6.2. Ordering Issues in a 20-Variable VAR

Next we provide a simple illustration of how the choice of ordering can influence empirical results in large VARs using a dataset of 20 popular U.S. monthly variables obtained from the FRED-MD database (McCracken and Ng 2016). The sample period is from 1959:03 to 2019:12. These variables, along with their transformations are listed in Online Appendix A. Four core variables, namely, industrial production, the unemployment rate, PCE inflation and the Federal funds rate, are ranked as the first to the fourth variables, respectively. The remaining 16 variables are ordered as in Carriero, Clark, and Marcellino (2019). We estimate the OI-VAR-SV and CS-VAR-SV models with the variables in this order (we call these models OI-VAR-SV-1 and CS-VAR-SV-1). Then we reverse the order of the variables and re-estimate them (these models are OI-VAR-SV-2 and CS-VAR-SV-2).

Estimates (posterior means) of reduced-form variances and covariances of selected variables are reported in Figures 8 and 9. That is, the first figure reports selected diagonal elements of $\mathbf{\Sigma}_t = \mathbf{B}_0^{-1}\mathbf{D}_t(\mathbf{B}_0^{-1})'$ under two different variable orderings, whereas the second figure reports selected off-diagonal elements of $\mathbf{\Sigma}_t$.

There are three points to note about these figures. First, as expected, estimates produced by the OI-VAR-SV model under the two different variable orderings are identical (up to MCMC approximation error). Second, estimates produced by the CS-VAR-SV model under the two different variable orderings are often similar to one another, but occasionally differ, particularly in periods of high volatility. This is in line with the results in Hartwig (2020), who shows that the reduced-form covariances in the application of Primiceri (2005) can be substantially different across alternative orderings of variables. Third, estimates from the CS-VAR-SV models are often similar to the OI-VAR-SV models, but sometimes are substantially different, particularly in the error covariances.

Estimates of other parameters that are not directly related to log-volatility can also be heavily influenced by the choice of ordering. For example, Table 2 reports the posterior means of the shrinkage hyperparameters κ_1 and κ_2 under the two models with two different orderings of the variables. As is clear from the table, the estimates of κ_1 and κ_2 can differ substantially when using the CS-VAR-SV model, depending on how the variables are ordered. For example, the estimate of κ_2 increases by 84% when one reverses the order of the variables. Of course, under the proposed order-invariant model the estimates are the same (up to MCMC approximation error).

We next present results relating to $\mathbf{B}_0$ which highlight the differences between our OI-VAR-SV-1 and the CS-VAR-SV-1.[9] Figure 10 reports the posterior means of $\mathbf{B}_0$ of these two models.

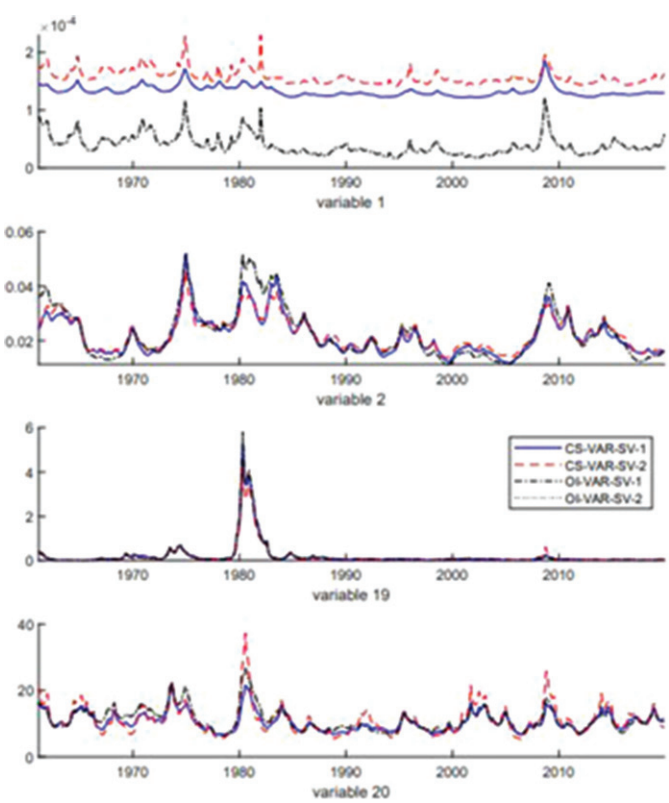

Figure 8. Estimates of time-varying reduced-form error variances from CS-VAR-SV and OI-VAR-SV under two different variable orderings.

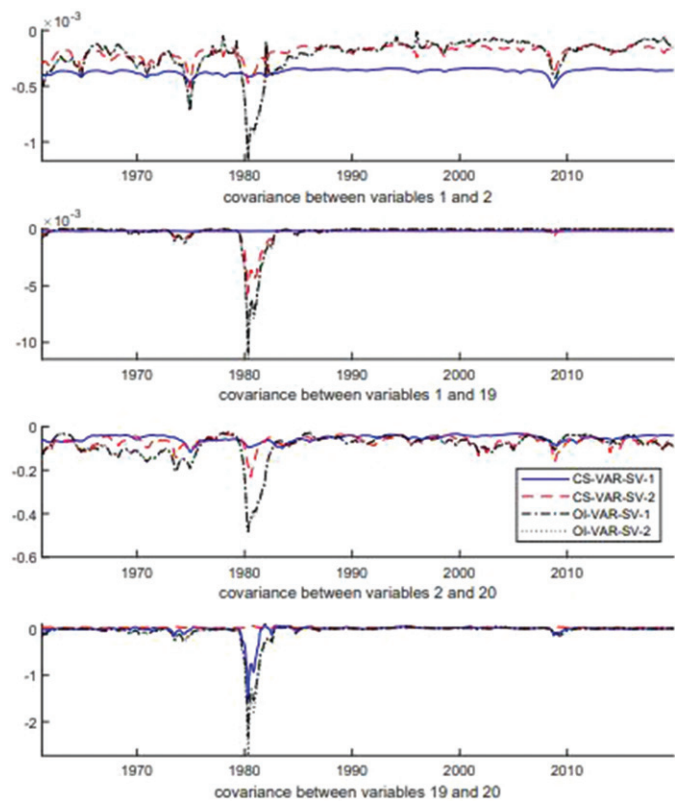

Figure 9. Estimates of time-varying reduced-form error covariances from from CS-VAR-SV and OI-VAR-SV under two different variable orderings.

Table 2. Posterior means of κ_1 and κ_2 from the proposed order-invariant stochastic volatility model under two different orderings (OI-VAR-SV-1 and OI-VAR-SV-2), as well as those from the model of Cogley and Sargent (2005) (CS-VAR-SV-1 and CS-VAR-SV-2).

	CS-VAR-SV-1	CS-VAR-SV-2	OI-VAR-SV-1	OI-VAR-SV-2
κ_1	0.0177	0.0170	0.0356	0.0358
κ_2	2.34×10^{-6}	4.30×10^{-6}	2.36×10^{-6}	2.25×10^{-6}

[9]Note that we are comparing the two models under the first ordering of the variables. The comparable figure using the second ordering reveals similar patterns.

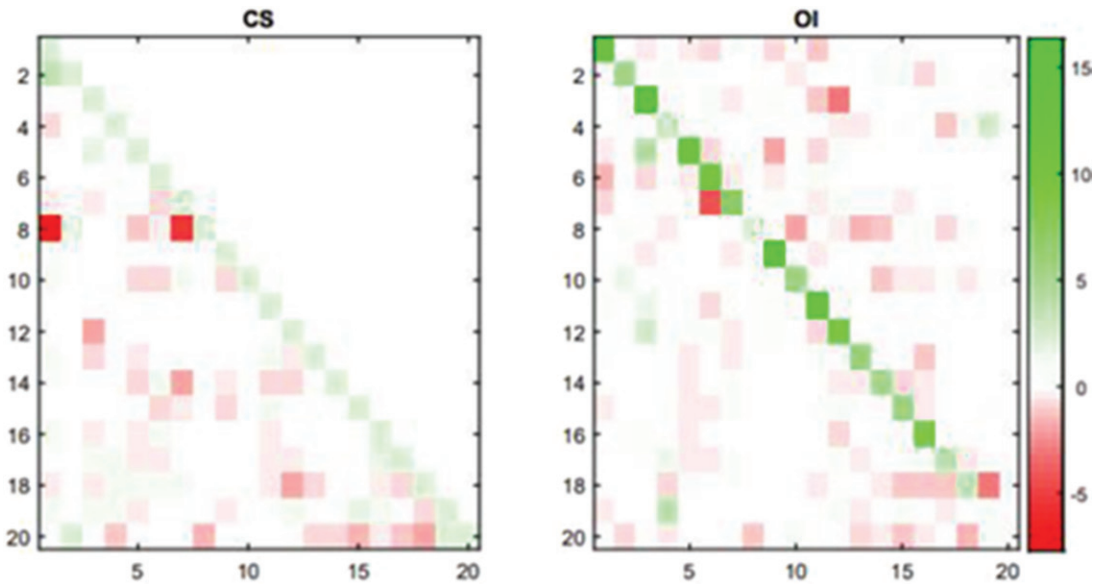

Figure 10. Posterior means of $\mathbf{B}_0$ using CS-VAR-SV-1 (left panel) and OI-VAR-SV-1 (right panel).

Note that the two panels of the figure are very different. Part of this difference is due to the fact that, under OI-VAR-SV, the SV processes are assumed to have zero unconditional means and the diagonal elements of $\mathbf{B}_0$ are unrestricted. Hence, the diagonal elements of $\mathbf{B}_0$ play a key role in adjusting the scale of the error variances. In contrast the diagonal elements of $\mathbf{B}_0$ are restricted to be one for the CS-VAR-SV model. But the key difference lies in the upper-triangular elements of $\mathbf{B}_0$. These are restricted to be zero in the CS-VAR-SV. But several of these upper triangular elements are estimated to be large in magnitude using our OI-VAR-SV model. The data strongly supports an unrestricted impact matrix $\mathbf{B}_0$. In the next section, we formally test the hypothesis that $\mathbf{B}_0$ is lower triangular via a Bayesian model comparison exercise.

6.3. Bayesian Model Comparison

In this section we conduct a series of model comparison exercises using the marginal likelihood. First, as highlighted in Herwartz and Lütkepohl (2014), in the presence of heteroscedasticity the lower triangular assumption on $\mathbf{B}_0$ is testable through the lens of the unrestricted OI-VAR-SV model. We test this lower triangular assumption and quantify the impact of different orderings. Another practical issue is the choice of m, the number of stochastic volatility processes, in the model. In the baseline OI-VAR-SV we set $m = n$, but it is plausible that the bias-variance trade-off might favor a smaller value of m. We therefore compare specifications with different values of m to provide guidance for practitioners.

The gold standard for Bayesian model comparison is the marginal likelihood or the marginal data density, defined for model M_k as

$$p(\mathbf{y} \mid M_k) = \int p(\mathbf{y} \mid \boldsymbol{\theta}_k, M_k) p(\boldsymbol{\theta}_k \mid M_k) \mathrm{d}\boldsymbol{\theta}_k,$$

where $\mathbf{y} = (\mathbf{y}_1', \ldots, \mathbf{y}_T')'$ is the full sample, $\boldsymbol{\theta}_k$ is the vector of model-specific parameters and latent variables, $p(\mathbf{y} \mid \boldsymbol{\theta}_k, M_k)$ is the likelihood and $p(\boldsymbol{\theta}_k \mid M_k)$ is the prior distribution. The integration involved in the marginal likelihood computation is typically very high dimensional for large-scale models. Despite recent advances, direct computation of the marginal likelihood for large VARs with stochastic volatility remains challenging. Here we circumvent this computational challenge by approximating the marginal likelihood using a sum of predictive likelihoods, which can be obtained by running the posterior sampler multiple times with an expanding window of data points. More specifically, note that we can factor the marginal likelihood as

$$p(\mathbf{y} \mid M_k) = p(\mathbf{y}_1 \mid M_k) \prod_{t=1}^{T-1} p(\mathbf{y}_{t+1} \mid \mathbf{y}_{1:t}, M_k),$$

where $\mathbf{y}_{1:t} = (\mathbf{y}_1', \ldots, \mathbf{y}_t')'$ denotes all the data up to time t and $p(\mathbf{y}_{t+1} \mid \mathbf{y}_{1:t}, M_k)$ is the predictive likelihood, that is, the one-step-ahead density forecast for $\mathbf{y}_{t+1}$ evaluated at the actual observation. Notice that in the above expression, the one-step-ahead density forecast $p(\mathbf{y}_1 \mid M_k)$ depends only on the prior distribution under model M_k. More generally, $p(\mathbf{y}_{t+1} \mid \mathbf{y}_{1:t}, M_k)$ is heavily influenced by the specific choice of the prior when t is small. To reduce the impact of the prior, one often discards part of the initial sample and uses instead the following approximation of the marginal likelihood:

$$p(\mathbf{y} \mid M_k) \approx \prod_{t=t_0}^{T-1} p(\mathbf{y}_{t+1} \mid \mathbf{y}_{1:t}, M_k).$$

We set t_0 so that the first 20 years are used as the training sample. Each predictive likelihood $p(\mathbf{y}_{t+1} \mid \mathbf{y}_{1:t}, M_k)$ is obtained by running a separate MCMC using data $\mathbf{y}_{1:t}$.

To test the lower triangular assumption on $\mathbf{B}_0$, as well as quantifying the impact of different orderings, we compare four models: OI-VAR-SV-1, OI-VAR-SV-2, CS-VAR-SV-1, and CS-VAR-SV-2. The log marginal likelihood estimates are reported in Table 3. Two points are immediately clear from these results. First, the data overwhelmingly favor the proposed OI-VAR-SV relative to the CS-VAR-SV. For instance, under the first ordering,

Table 3. Log marginal likelihood values of the proposed order-invariant stochastic volatility model under two different orderings (OI-VAR-SV-1 and OI-VAR-SV-2), as well as those from the model of Cogley and Sargent (2005) (CS-VAR-SV-1 and CS-VAR-SV-2).

	CS-VAR-SV-1	CS-VAR-SV-2	OI-VAR-SV-1	OI-VAR-SV-2
log ML	−26,301	−26,709	−22,714	−22,712

Table 4. Log marginal likelihood values of the OI-VAR-SV-1 with m stochastic volatility processes.

m	17	18	19	20
log ML	−22,713	−22,701	−22,684	−22,714

the difference in marginal likelihood values is over 3500 in log scale. Second, the marginal likelihood values of the proposed OI-VAR-SV under the two different orderings are the same (up to MCMC approximation error), whereas for the CS-VAR-SV, the two orderings give very different marginal likelihood values—a difference of more than 400 in log scale. All in all, these results further highlight the advantages of the proposed OI-VAR-SV: not only is it robust to different variable orderings, it is also more flexible and provides a much better fit of the data.

Next, we investigate the number of SV processes m needed to fit the data well, while accounting for model complexity. To that end, we compare the baseline OI-VAR-SV-1 with different values of m.[10] When $m \leqslant n-2=18$, the impact matrix $\mathbf{B}_0$ is only partially identified, but the log marginal likelihood values remain valid. The results in Table 4 show that the data favor the specification with $m = 19$ SV processes. Interestingly, it outperforms both a more restricted version ($m = 18$) as well as a more flexible one ($m = 20$), highlighting the value of conducting a model comparison exercise to select m. In addition, that the best value of m is large suggests that the co-volatility structure in macroeconomic data is complex and requires more a flexible modeling approach.

7. A Forecasting Exercise

7.1. Comparison of OI-VAR-SV to CS-VAR-SV

Since VARs are commonly used for forecasting, it is interesting to investigate how the sensitivity to ordering of the CS-VAR-SV models affects forecasts and whether working with our order-invariant specification can lead to forecast improvements. Accordingly we carry out a forecasting exercise using the monthly macroeconomic dataset of Section 6.2 for the four models (OI-VAR-SV-1, OI-VAR-SV-2, CS-VAR-SV-1, and CS-VAR-SV-2). Forecast performance of the models is evaluated from 1970:03 till the end of the sample. Root mean squared forecast errors (RMSFEs) are used to evaluate the quality of point forecasts and averages of log predictive likelihoods (ALPLs) are used to evaluate the quality of density forecasts. We focus on four core variables: industrial production, the unemployment rate, PCE inflation and the Federal funds rate. Results for three different forecast horizons ($h = 1, 6, 12$) are presented in Table 5. We use the iterated method of forecasting to compute longer horizon forecasts.

Note first that, as expected, OI-VAR-SV-1 and OI-VAR-SV-2 are producing the same forecasts (up to MCMC approximation error). CS-VAR-SV-1 and CS-VAR-SV-2 are often producing forecasts which are substantially different both from one another and from those produced by the order-invariant approaches. These differences are not that noticeable in the RMSFEs, but are very noticeable in the ALPLs. This suggests ordering issues are more important for density forecasts than for point forecasts. This is not surprising since ordering issues are important for the multivariate stochastic volatility process (which plays an important role in determining predictive variances) but not for the VAR coefficients (which are the key determinants of predictive means). These findings are similar to those found in smaller VAR-SVs by Arias, Rubio-Ramirez, and Shin (2022).

A careful examination of the ALPL results indicates that the best forecasting model is almost always the OI-VAR-SV and in most cases the forecast improvements are statistically significant relative to the CS-VAR-SV-1 benchmark. A comparison of CS-VAR-SV-1 to CS-VAR-SV-2 yields no simple conclusion. Remember that CS-VAR-SV-1 uses a similar ordering of Carriero, Clark, and Marcellino (2019). For most variables and forecast horizons, this ordering is leading to higher ALPLs than the reverse ordering. This finding is also consistent with the results in Arias, Rubio-Ramirez, and Shin (2022). But there are exceptions such as forecasting the Fed funds rate for $h = 6$ and $h = 12$. But the important point is not that, when using CS-VAR-SV models, one ordering is better than the other. The important points are that ordering matters (often in a statistically significant way) and that different variables prefer different orderings. That is, there is no one ordering that forecasts all of the variables best.

7.2. Adding Time Variation to $\mathbf{B}_0$

So far all the models in this article have assumed $\mathbf{B}_0$ to be constant over time and we have used the specification of Cogley and Sargent (2005) which maintains this assumption as a benchmark in our investigation of ordering issues. However, following Primiceri (2005) many papers have allowed for time variation in $\mathbf{B}_0$ as well. It is straightforward to extend our model to allow for this, leading to the order-invariant time varying parameter VAR-SV, or OI-TVP-VAR-SV which replaces $\mathbf{B}_0$ by $\mathbf{B}_{0,t}$ and assumes the elements of the latter follow random walks. The definition of the OI-TVP-VAR-SV along with an MCMC algorithm which allows for Bayesian inference is given in the Online Appendix. The Online Appendix also shows that the model of Primiceri (2005), which uses a lower triangular specification for $\mathbf{B}_{0,t}$, is subject to the same sensitivity to ordering issues as in the model of Cogley and Sargent (2005).

One issue with any TVP-VAR is that the computational burden can be very high in large TVP-VARs when using MCMC. But a lack of identification of a model does not preclude good forecast performance. Accordingly, we have repeated our forecasting exercise using a OI-TVP-VAR-SV involving our four core variables as well as the TVP-VAR-SV which assumes a lower triangular form for $\mathbf{B}_{0,t}$. We consider two versions of the

[10]For $m < n$, the $n \times n$ diagonal matrix $\mathbf{D}_t$ in (2) is set to be $\mathbf{D}_t = \text{diag}(e^{h_{1,t}}, \ldots, e^{h_{m,t}}, 1, \ldots, 1)$.

Table 5. RMSFE and ALPL of four core macroeconomic time series.

Variables	Models	RMSFE			ALPL		
		$h = 1$	$h = 6$	$h = 12$	$h = 1$	$h = 6$	$h = 12$
Industrial production	CS-VAR-SV-1	0.007	**0.007**	**0.007**	2.071	2.159	2.266
	CS-VAR-SV-2	0.007**	0.007**	0.007**	1.522***	1.651***	1.767***
	OI-VAR-SV-1	**0.007**	0.007	0.007***	**3.660*****	3.458***	**3.360*****
	OI-VAR-SV-2	0.007	0.007	0.007***	3.659***	**3.459*****	3.358***
Unemployment rate	CS-VAR-SV-1	0.159	0.169	0.173	−0.005	−0.689	−0.707
	CS-VAR-SV-2	0.158	0.169	0.173	−0.124	−0.761	−0.742
	OI-VAR-SV-1	**0.158**	**0.167**	**0.173**	**0.463*****	**0.276**	0.152
	OI-VAR-SV-2	0.158	0.167*	0.173	0.463***	0.276	**0.156**
PCE inflation	CS-VAR-SV-1	0.002	**0.002**	**0.002**	2.210	2.311	2.425
	CS-VAR-SV-2	0.002***	0.002	0.002	1.811***	1.919***	2.024***
	OI-VAR-SV-1	**0.002****	0.002	0.002*	4.821***	4.448***	4.249***
	OI-VAR-SV-2	0.002**	0.002	0.002	**4.823*****	**4.453*****	**4.260*****
Federal funds rate	CS-VAR-SV-1	**0.492**	**1.566**	**2.216**	0.222	−8.473	−16.704
	CS-VAR-SV-2	0.497	1.597*	2.264	−0.327***	**−6.708*****	**−12.474*****
	OI-VAR-SV-1	0.493	1.576	2.233	0.294***	−6.956*	−13.466*
	OI-VAR-SV-2	0.494	1.570	2.236	**0.296*****	−7.039	−13.662

NOTE: The bold figure indicates the best model in each case. *, **, and *** denote, respectively, 0.10, 0.05, and 0.01 significance level for a two-sided Diebold and Mariano (1995) test. The benchmark model in the Diebold Mariano test is CS-VAR-SV-1.

Table 6. RMSFE and ALPL of four core macroeconomic time series.

Variables	Models	RMSFE			ALPL		
		$h = 1$	$h = 6$	$h = 12$	$h = 1$	$h = 6$	$h = 12$
Industrial production	OI-TVP-VAR-SV	0.007***	0.008*	0.008	3.465***	2.673***	2.097***
	Primiceri-1	0.009	0.022	0.073	1.757	−0.468	−3.695
	Primiceri-2	0.052***	0.966***	6.217***	−0.344***	−2.685***	−6.085***
Unemployment rate	OI-TVP-VAR-SV	0.203***	0.207*	0.200*	0.249*	−0.839***	−1.697***
	Primiceri-1	0.175	0.194	0.252	0.339	−3.120	−6.878
	Primiceri-2	0.548**	1.304***	5.200***	−1.046***	−5.199***	−9.446**
PCE inflation	OI-TVP-VAR-SV	0.002	0.002	0.002	4.234***	3.842***	3.586
	Primiceri-1	0.009	0.036	0.117	1.573	−0.832	−3.874
	Primiceri-2	0.054**	0.618***	4.549**	−0.222***	−2.576***	−6.042***
Fed funds rate	OI-TVP-VAR-SV	0.596***	1.998	2.692	−0.298	−8.676***	−15.755**
	Primiceri-1	0.529	1.808	3.905	0.048	6.245	−11.650
	Primiceri-2	0.549	4.109***	21.992***	0.058***	−8.501***	−22.344***

NOTE: *, ** and *** denote, respectively, 0.10, 0.05, and 0.01 significance level for a two-sided Diebold and Mariano (1995) test. The benchmark model for the Diebold-Mariano test is Primiceri-1.

latter with the variables ordered in two different ways (i.e., in the order they appear in Table 5 and in the reverse order. These are labeled Primiceri-1 and Primiceri-2 in the table). Table 6 provides the results and indicates that our OI-TVP-VAR-SV is typically forecasting better than TVP models which use the lower triangular form. The importance of ordering can be seen in the non-order invariant TVP-VAR-SV in that Primiceri-2 is typically forecasting much worse than Primiceri-1.

It is also interesting to compare Table 5 (involving 20 dimensional VARs without TVPs) to Table 6 (involving four variable TVP-VARs). The pattern is mixed, but in most cases the high-dimensional OI-VAR-SV is beating its lower-dimensional TVP counterpart. This indicates that the extra variables in the larger model are typically helping improve forecasts more than simply adding TVPs to a smaller model.

8. Conclusions

In this article, we have demonstrated, both theoretically and empirically, the consequences of working with VAR-SVs which use a lower triangular parameterization of the error covariance matrices such as that used in Cogley and Sargent (2005) and are thus not order invariant. We have proposed a new specification which is order invariant which involves working with an unrestricted version of the impact matrix, $\mathbf{B}_0$. Such a model would be unidentified in the homoscedastic VAR but we draw on Bertsche and Braun (2022) to establish that the incorporation of SV identifies the model. We develop an MCMC algorithm which allows for Bayesian inference and prediction in our order-invariant model. In an empirical exercise involving 20 macroeconomic variables we demonstrate the ability of our methods to produce accurate forecasts in a computationally efficient manner.

Supplementary Materials

The supplementary materials contain a description of the dataset, technical details on the estimation procedures, and additional analytical and simulation results.

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References

Archakov, I., and Hansen, P. (2021), "A New Parametrization of Correlation Matrices," *Econometrica*, 89, 1699–1715. [826]

Arias, J. E., Rubio-Ramirez, J. F., and Shin, M. (2022), "Macroeconomic Forecasting and Variable Ordering in Multivariate Stochastic Volatility Models," *Journal of Econometrics*, forthcoming. [825,826,835]

Asai, M., and McAleer, M. (2009), "The Structure of Dynamic Correlations in Multivariate Stochastic Volatility Models," *Journal of Econometrics*, 150, 182–192. [826]

Banbura, M., Giannone, D., and Reichlin, L. (2010), "Large Bayesian Vector Auto Regressions," *Journal of Applied Econometrics*, 25, 71–92. [825]

Bertsche, D., and Braun, R. (2022), "Identification of Structural Vector Autoregressions by Stochastic Volatility," *Journal of Business & Economic Statistics*, 40, 328–341, DOI: 10.1080/07350015.2020.1813588. [825,827,836]

Bognanni, M. (2018), "A Class of Time-Varying Parameter Structural VARs for Inference Under Exact or Set Identification," FRB of Cleveland working paper. [825,826]

Carriero, A., Chan, J. C. C., Clark, T. E., and Marcellino, M. G. (2022), "Corrigendum To: Large Bayesian Vector Autoregressions with Stochastic Volatility and Non-conjugate Priors," *Journal of Econometrics*, 227, 506–512. [831]

Carriero, A., Clark, T. E., and Marcellino, M. G. (2015), "Bayesian VARs: Specification Choices and Forecast Accuracy," *Journal of Applied Econometrics*, 30, 46–73. [829]

——— (2016), "Common Drifting Volatility in Large Bayesian VARs," *Journal of Business and Economic Statistics*, 34, 375–390. [826]

——— (2019): "Large Bayesian Vector Autoregressions with Stochastic Volatility and Non-conjugate Priors," *Journal of Econometrics*, 212, 137–154. [825,826,831,833,835]

Carvalho, C. M., Polson, N. G., and Scott, J. G. (2010), "The Horseshoe Estimator for Sparse Signals," *Biometrika*, 97, 465–480. [829]

Chan, J. C. C. (2020), "Large Bayesian VARs: A Flexible Kronecker Error Covariance Structure," *Journal of Business and Economic Statistics*, 38, 68–79. [826]

——— (2021), "Minnesota-Type Adaptive Hierarchical Priors for Large Bayesian VARs," *International Journal of Forecasting*, 37, 1212–1226. [828]

——— (2022): "Large Hybrid Time-Varying Parameter VARs," *Journal of Business and Economic Statistics*, forthcoming. [825]

Chan, J. C. C., Doucet, A., León-González, R., and Strachan, R. W. (2018), "Multivariate Stochastic Volatility with Co-heteroscedasticity," CAMA working paper. [825,826]

Clark, T. E. (2011), "Real-Time Density Forecasts from Bayesian Vector Autoregressions with Stochastic Volatility," *Journal of Business and Economic Statistics*, 29, 327–341. [825]

Cogley, T., and Sargent, T. J. (2005), "Drifts and Volatilities: Monetary Policies and Outcomes in the Post WWII US," *Review of Economic Dynamics*, 8, 262–302. [825,826,827,831,833,835,836]

Diebold, F. X., and Mariano, R. S. (1995), "Comparing Predictive Accuracy," *Journal of Business and Economic Statistics*, 13, 253–263. [836]

Hartwig, B. (2020), "Robust Inference in Time-Varying Structural VAR Models: The DC-Cholesky Multivariate Stochastic Volatility Model," Deutsche Bundesbank discussion paper no 34/2020. [825,833]

Herwartz, H., and Lütkepohl, H. (2014), "Structural Vector Autoregressions with Markov Switching: Combining Conventional with Statistical Identification of Shocks," *Journal of Econometrics*, 183, 104–116. [834]

Karlsson, S. (2013), "Forecasting with Bayesian Vector Autoregressions," in *Handbook of Economic Forecasting* (Vol. 2), eds. G. Elliott and A. Timmermann, pp. 791–897, Amsterdam: Elsevier. [829]

Kastner, G. (2019), "Sparse Bayesian Time-Varying Covariance Estimation in Many Dimensions," *Journal of Econometrics*, 210, 98–115. [826]

Kaufmann, S., and Schumacher, C. (2019), "Bayesian Estimation of Sparse Dynamic Factor Models with Order-Independent and Ex-post Mode Identification," *Journal of Econometrics*, 210, 116–134. [827]

Lanne, M., Lütkepohl, H., and Maciejowska, K. (2010), "Structural Vector Autoregressions with Markov Switching," *Journal of Economic Dynamics and Control*, 34, 121–131. [827]

Leeper, E. M., Sims, C. A., and Zha, T. (1996), "What Does Monetary Policy Do?" *Brookings Papers on Economic Activity*, 1996, 1–78. [825]

Levy, B., and Lopes, H. (2021), "Dynamic Ordering Learning in Multivariate Forecasting," *https://arxiv.org/abs/2101.04164*. [825]

Lewis, D. J. (2021), "Identifying Shocks via Time-Varying Volatility," *The Review of Economic Studies*, forthcoming. [827]

Makalic, E., and Schmidt, D. F. (2016), "A Simple Sampler for the Horseshoe Estimator," *IEEE Signal Processing Letters*, 23, 179–182. [829]

McCracken, M. W., and Ng, S. (2016), "FRED-MD: A Monthly Database for Macroeconomic Research," *Journal of Business and Economic Statistics*, 34, 574–589. [833]

Philipov, A., and Glickman, M. (2006), "Multivariate Stochastic Volatility via Wishart Processes," *Journal of Business and Economic Statistics*, 24, 313–328. [826]

Prado, R., and West, M. (2010), *Time Series: Modeling, Computation, and Inference*, Boca Raton, FL: Chapman and Hall/CRC. [826]

Primiceri, G. E. (2005), "Time Varying Structural Vector Autoregressions and Monetary Policy," *Review of Economic Studies*, 72, 821–852. [825,826,827,831,833,835]

Rigobon, R. (2003), "Identification through Heteroskedasticity," *Review of Economics and Statistics*, 85, 777–792. [827]

Shin, M., and Zhong, M. (2020), "A New Approach to Identifying the Real Effects of Uncertainty Shocks," *Journal of Business and Economic Statistics*, 38, 367–379. [826]

Sims, C. A. (1980), "Macroeconomics and Reality," *Econometrica*, 48, 1–48. [825]

Uhlig, H. (1997), "Bayesian Vector Autoregressions with Stochastic Volatility," *Econometrica*, 65, 59–73. [826]

Villani, M. (2009), "Steady-State Priors for Vector Autoregressions," *Journal of Applied Econometrics*, 24, 630–650. [829,830]

Waggoner, D., and Zha, T. (2003), "A Gibbs Sampler for Structural Vector Autoregressions," *Journal of Economic Dynamics and Control*, 28, 349–366. [829]

West, M., and Harrison, J. (2006), *Bayesian Forecasting and Dynamic Models*, New York: Springer. [826]

Wu, P., and Koop, G. (2022), "Estimating the Ordering of Variables in a VAR Using a Plackett-Luce Prior," manuscript. [825]